

Boris Kafidov

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EDUCATION

University of Toronto

Honours B.Sc. in Computer Science & Mathematics, Focus in Artificial Intelligence

Toronto, ON

Sep. 2024 – Jun. 2028

EXPERIENCE

Amazon

Jr. Software Development Engineer (Internship)

Toronto, ON

Jun. 2026 – Present

- Built and launched Amazon Books' mobile format-selector redesign in TypeScript across 11 marketplaces on web, Android, and iOS; analyzed a 1.89M-customer A/B test projecting \$3.37M in additional annual sales
- Designed a legacy-to-microfrontend migration of a related-book widget on pages averaging 14.3M weekly visits, reusing a shared API-backed carousel and avoiding another component to deploy and maintain
- Fixed series-description errors across three regions by correcting API inputs, preserving the request contract
- Resolved a rendering race between Web Components and a legacy container that hid descriptions on Fire tablets

University of Toronto, Wheeler Lab

Software Engineer (Work Study)

Toronto, ON

Oct. 2025 – Mar. 2026

- Batched microscope setting changes in Python/PySide6, preserving dependency order and waiting once for all changes; added sequential fallback on timeout and timing logs for camera, stage, and channel operations
- Enabled development of a Linux microscope application without connected hardware by simulating the camera and stage; fixed a Qt signal race that caused unnecessary 10-second waits during tiled imaging test runs
- Integrated serial and USB HID microscope light sources behind one controller; added automatic backend selection, binary commands with CRC32 checksums, OS-specific message formats, bounded reads, and safe shutdown

RESEARCH

Log-concavity of elementary coefficients for low-rank abelian Hessenberg graphs

Aug. 2026

Sole-authored arXiv preprint; under review at *Annals of Combinatorics*. Disproves the general log-concavity conjecture

Dittert's conjecture in dimension 16 via a joint-deficit scaling lemma

Jul. 2026

Sole-authored arXiv preprint. Proves dimension 16 via matrix scaling and permanent bounds, settling all $n \geq 16$

PROJECTS

CreditCombo | Live App | JavaScript, Web Workers, Cloudflare Workers

Feb. 2026 – Mar. 2026

- Independently built and launched a browser app that recommends credit-card combinations and which card to use for each spending category, modeling annual rewards after fees across 151 Canadian cards and 51 rewards programs
- Built a constrained optimization engine that prunes the combinatorial search space and assigns users' spending to cards by marginal return, accounting for shared rewards caps to avoid overstating projected returns
- Ran portfolio search in a Web Worker to keep the interface responsive; reused results through a bounded LRU cache and discarded outdated search responses to prevent stale recommendations after user inputs changed

APA Site Choice Transformer | Python, PyTorch, Hugging Face, GCP

Sep. 2025

- Co-led a five-person team to second place at the Toronto Bioinformatics Hackathon (\$500 award)
- Fine-tuned DNABERT-2 (117M parameters) to predict alternative polyadenylation sites from approximately 800k labeled sequence windows on GCP-hosted A100 GPUs; the team's model achieved 0.93 AUROC and 0.94 AUPRC
- Built a PyTorch pipeline with mixed precision, gradient accumulation, and chromosome holdout support; prevented GPU out-of-memory errors by moving evaluation logits to CPU and added a batched evaluator for saved models

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, Java, C++, C

Frameworks/Libraries: PyTorch, Hugging Face Transformers, PySide6, scikit-learn, NumPy, Pandas, Node.js

Platforms/Tools: Linux, Git, AWS CDK, CloudWatch, GCP, Cloudflare Workers, Bash

Areas: Machine Learning, Deep Learning, Algorithms & Data Structures, Concurrency, Performance & Reliability